Part 1 briefing and video script

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What the team did in Part 1 and why, in plain words, followed by a timed script for the Part 1 segment of the video.

1. EDA: what the data showed

The exploration found three kinds of problem: recording errors to fix, a target that is immature and heavy-tailed, and variables that carry information 2025 policies will not have. Both pipelines found largely the same issues on their own, which is a useful cross-check. The data: 2,409,311 training rows (2021 to 2024), 163,150 claims and 626,639 test rows.

What we saw	Evidence	What it means, what we did
Exposure in mixed units	3,000 to 4,000 rows per line above one year: values 1.0139 (a 360-day basis) and 365 (a day count)	Exposure is the denominator of every frequency, so it is recomputed from the policy dates
Sums insured 1,000 times too large	About 1,400 rows per home line above 5 million, in train and test, exactly 1,000 times the same policy's other years	Divided by 1,000 and flagged
Inconsistent construction codes	11 raw spellings, including the codes T and W (about 6% each in train, only W in test)	Lower-cased; T and W both mapped to timber
Motor customer IDs lost their leading zeros	About 98,000 motor rows failed to join the customer table	Padded to 8 characters: 100% now join
About 600 duplicate claims	Same policy, date, peril and amount, different claim ID	Removed, otherwise their cost counts twice

What we saw	Evidence	What it means, what we did
Claims carry no policy row	Each claim has to be placed in a policy period	Matched by loss date: 99.7% match, the rest mostly December 2020 losses; total cost reconciles exactly
Recent claims not yet reported (IBNR)	Motor 2024: only 77% of ultimate cost reported (pure premium 454 reported, 585 ultimate). Bodily injury is 43% of motor cost and only 14% of it is reported at 30 days	Training on raw 2024 cost would underprice, so claims are developed (section 2)
Heavy tails	One open bodily-injury reserve of 11.9 million; 8.8% of motor cost sits above the 99.5th percentile (4.2% buildings, 2.1% contents)	A few claims would drive the rating factors, so claims are capped with a load
Rising cost	Ultimate pure premium 2021 to 2024: buildings 384 to 481, contents 106 to 121, motor 564 to 585	A trend is needed to reach 2025
A leaking variable	`customer_service_contacts` averages 3.3 for claimants against 0.4 for others; 2025 rows average 0.40, like non-claimants	Contacts are caused by claims, so the variable is dropped
Claim history unknowable for 2025	For a 2025 row, the 12 months before its start fall mostly after the data cut-off: motor rows starting in Q4 with a prior claim are 4.0% in training but 0.4% in test	Only claims known on 1 January of the row's year are counted (Oussama's Improved 3)
An older 2025 book	Median renewal count falls from 6 to 3, vehicle age rises from 7 to 9 years, NCB from 7 to 8 years	Consistent drift; checked, no fix needed

The weather variables also identify the year almost perfectly: a classifier separates 2024 from 2025 rows with AUC 0.999. Oussama removed them from LightGBM to test this, then restored them when the score did not change.

2. Data preparation: a target that means the right thing

The target is the ultimate claim cost of each row, so most of the preparation turns a snapshot of claims at the end of 2024 into that quantity. Every fix is applied in code when the data is loaded; the raw files are never edited, and Yichuan's repo checks them with a SHA256 test.

1. **Clean the policy tables.** Exposure from the dates, sums insured divided by 1,000, construction codes unified, motor IDs padded (section 1).
2. **Build the claim cost.** Cost is incurred minus recoveries, repudiated claims and duplicates excluded. Each claim goes to the row whose period contains its loss date. Incurred is treated as already net of the excess: 20 to 27% of claims sit below the policy excess, which only makes sense after it.
3. **Allow for unreported claims (IBNR).** Losses from 2021 are fully reported by the end of 2024, so they show what share of claims is reported t days after the loss. Averaging that share over a row's days at risk gives its completeness: about 1.0 for 2021, 0.77 for motor 2024 on the cost-weighted curve. Models then compare cost with exposure × completeness, not with full exposure.
    - Why this way: the observed claims stay as observed, real counts and amounts, so the Poisson model stays valid, while the fitted rate is an ultimate rate. It is the same as dividing reported cost by completeness.
    - The one difference between the pipelines: Oussama's curve counts claims, Yichuan's weights them by cost. In motor the late claims are the expensive bodily-injury ones, so the cost-weighted curve gives a larger and more accurate IBNR for a cost target. Expect this question.
4. **Cap large losses.** Each claim is capped at its line's 99.5th percentile (motor 51,585, buildings 147,445, contents 28,449). The cost above the cap comes back as a multiplicative load per line, total cost divided by capped cost. Why: rating factors should explain everyday claims, and one random 11.9 million reserve would otherwise move a whole region's factor. The load keeps the total premium right.
5. **Project to 2025.** Both use a log-linear trend, but on different things.
    - Oussama: trend in average severity by policy year, kept for a model only if it improved its 2024 score (always on for CatBoost).
    - Yichuan: trend in developed pure premium per line. The public board preferred half of it (0.12464) to none (0.12376) or all of it (0.12453), so predictions use the square root of the factor.

- Known risk: written premium and sum insured already rise every year and enter the models, so part of the trend may be counted twice (audit finding B4). A probe with a quarter of the trend scored lower, so half was kept.

3. Features and variable selection

Features come from joining the customer, address and weather tables to each policy row. The main decisions were about leakage first, then about which variables the GLM keeps.

Added: customer age, years as a customer, building age, log sum insured, the other lines the customer holds (`holds_motor`, `n_lines_held`), and the customer's earlier claims.

Removed, and why:

- `customer_service_contacts`: caused by claims, so it leaks the answer.
- `renewal_decision`: only known after the period ends.
- Surname, date of birth, email domain: noise, or a copy of age.
- Yichuan's per-line drop list (C026): `branch_id`, `previous_insurer`, motor `make_model`, cross-line history and household counts. They carried no frequency signal and the trees were overfitting them.

The claim-history fix. A claim only counts if it was reported before 1 January of the row's year. For 2025 rows that date is the data cut-off, so training and test see the same kind of information, the information an underwriter has at quotation. The 2024 validation could not detect this problem because 2024 history was complete; only the leaderboard showed it.

How the GLM chooses its variables (Oussama). A forward selection, run separately for frequency and severity on each line:

1. Numeric variables are cut into up to 10 quantile bins; levels under 0.5% of exposure are grouped as "other".
2. Starting from the intercept, every candidate is ranked by a score test. It needs only the current fit, so 62 to 72 candidates cost one fit, not 72.
3. The best three are refitted and the one that lowers BIC most is kept; selection stops when none lowers it. BIC charges log(n) per parameter, about 13 on 500,000 rows, against 2 for AIC, so it keeps only variables with a real effect.

Line	Frequency variables	Severity variables	First frequency picks, in order
Motor	23	3	written premium, cover type, years licensed, annual mileage, voluntary excess, NCB years
Buildings	16	3	building age, flood zone, accidental damage cover, prior claims (5 years), construction type
Contents	14	4	accidental damage cover, flood zone, voluntary excess, policyholder age, prior claims (5 years)

Severity keeps only 3 or 4 variables on every line: given the policy, claim size is mostly random, and the rating signal sits in frequency. That is a good actuarial point to make in the video. Written premium is the strongest motor variable because it carries the insurer's current tariff; a judge may note that the model partly learns the existing price.

The other models. The GAM uses the GLM's variables, with numeric ones as penalised splines (P-splines) and one smoothing parameter chosen by AIC. The trees take every variable that does not leak. LightGBM gets categoricals as integer codes, because its own categorical splits overfit `make_model`, which has 300 levels. CatBoost encodes them with ordered target statistics.

4. Modelling: two pipelines, then one blend

The two pipelines answer the same question in different ways: Oussama splits frequency and severity, Yichuan fits pure premium directly. The Part 1 method in the report and video is a 0.7 / 0.3 blend of the two, as the team lead decided on 30 October.

	Oussama (Blend4)	Yichuan (S09)
What each model predicts	GLM and GAM: frequency (Poisson) × severity (Gamma). Trees: cost directly (Tweedie 1.5)	Every model: capped cost per unit of exposure (Tweedie 1.5)
Models	GLM, GAM, LightGBM, CatBoost	Banded GLM, spline GAM, LightGBM, XGBoost, CatBoost

	Oussama (Blend4)	Yichuan (S09)
Fitted per	Line of business	Line of business
IBNR	Exposure × count-based completeness	Exposure × cost-weighted completeness
Variable choice	Forward selection on BIC (GLM, GAM); every non-leaking variable (trees)	Per-line drop list (C026); tuned settings per line
Tuning rule	Random search over two validation years (LightGBM); three settings (CatBoost)	Set in advance: adopt a change only for a mean gain of at least 0.002 with no fold lost
Blend weights	Equal: fitted weights scored lower (public 0.12636 against 0.12735)	0.05 grid, kept only if they beat equal weights on both folds; motor stayed equal
Level	Large-loss load × severity trend	Rescaled to the training total, large-loss load, half trend
Final refit	2021 to 2024	2021 to 2024, tree models averaged over 3 seeds

Why frequency × severity. It separates how often claims happen from how much they cost, and gives multiplicative factors an actuary can read and challenge. That is the classic tariff structure.

Why pure premium with Tweedie. It models the mass of zero-claim policies and the cost in one step, with the same Tweedie power as the metric. Yichuan tested a per-peril frequency × severity model (C023): it scored 0.0801 against 0.0893 for the best single motor model, so pure premium won there.

Why GLM and GAM next to trees. The GLM is the readable, auditable tariff. The GAM keeps that structure but lets each continuous effect take its own shape. The trees find interactions the GLM cannot. Models with different structures make different errors, so averaging them cuts variance.

The combined Part 1 method. Blend4 is first re-levelled to S09's total on each line, then the prediction is 0.7 × S09 + 0.3 × Blend4. Re-levelling means the blend changes the ranking of policies within a line, not the line totals. It works because the two pipelines

correlate at 0.965: high, but different enough to correct each other. The weight 0.3 keeps the ranked entry as the larger part; 0.5 scored slightly lower (private 0.13707 against 0.13718).

5. Validation and results

The Part 1 method scores 0.13718 on the private board, against 0.13689 for the ranked entry S09 (rank 12 of 51) and 0.13923 for the winner. Both pipelines validate on a later year, never on a random split.

Why a later year. Pricing means predicting a future year, so the test must too. A random split would put the same policy's 2022 and 2023 rows on both sides and flatter the score. Oussama fits on 2021 to 2023 and scores 2024; Yichuan uses two expanding folds, 2021–22 to 2023 and 2021–23 to 2024.

Comparing like with like. 2024 is immature, so reported 2024 cost is scored against prediction × completeness, never against the full ultimate prediction. Yichuan's `d2_cal` also rescales each fold to its actual total, so it measures ranking, not level.

Oussama's 2024 validation, Tweedie D^2 (power 1.5):

Model	All lines	Buildings	Contents	Motor
GLM	0.1296	0.1071	0.0872	0.0834
GAM	0.1306	0.1094	0.0890	0.0831
LightGBM	0.1348	0.1157	0.0935	0.0857
CatBoost	0.1344	0.1147	0.0932	0.0858
Blend4	**0.1351**	0.1156	**0.0940**	**0.0865**

Trees beat the GLM by about 0.005, and the blend is best overall. CatBoost's score is slightly optimistic, because its tree count was chosen with early stopping on 2024.

Kaggle scores:

Entry	Public	Private
S01: Yichuan's first equal blend	0.12453	0.13347
S06: per-line blend weights	0.12579	0.13505

Entry	Public	Private
S09: Yichuan's final, ranked 12 of 51	0.12863	0.13689
Blend4: Oussama's final	0.12736	0.13526
0.7 S09 + 0.3 Blend4 (late submission, not ranked)	0.12949	0.13718

What moved the scores.

- Yichuan: per-line blend weights (+0.0010, S06); the feature drop list with CatBoost and early stopping on an inner sample (+0.0027, S07); seed averaging (+0.0001, S09).
- Oussama: fixing LightGBM's starting point (every policy started at 1 per year instead of about 300, so the trees wasted effort on the overall level); the claim-history fix; adding CatBoost. Public score rose from 0.12592 to 0.12758.

The lesson worth saying aloud. Two problems were invisible in validation and only the leaderboard showed them: the claim history (complete in 2024, not in 2025) and LightGBM alone ranking first on 2024 but scoring lowest of three files on the board (0.12272 against 0.12742 for the blend). Validation has to reproduce the information available when the price is set.

Why a D^2 of 0.13 is not weak. D^2 compares the model with one constant rate for everyone. At policy level most cost is pure chance, since most policies have no claim, so even a perfect tariff explains a small share. The private board ran about 0.008 above the public board for every file.

6. Video script: the Part 1 segment

The team script in `docs/VIDEO_SCRIPT.md` already runs 4:48 of the 5:00, at 135 words per minute, with Part 1 in five lines (94 words) of section 1. So there are two versions: a fuller segment if the team makes room, and a drop-in exactly as long as the lines it replaces. On-screen cues use the dashboard panels named in the team script.

Version A: a dedicated Part 1 segment (210 words, about 1:33)

On screen	Spoken
"Data quality: ten documented issues"	In Part 1, we predict the ultimate 2025 claim cost of every motor, buildings and contents policy.

On screen	Spoken
(same)	We started with a data audit: exposure in mixed units, sums insured a thousand times too large and 600 duplicate claims, all fixed in code.
`docs/target_summary.txt`, motor block: completeness by year	Claims stop at the end of 2024, so only 77 percent of 2024 motor cost is reported. Bodily injury is slowest: 14 percent at 30 days.
(same)	We develop it with a reporting curve from 2021 losses, and cap claims at the 99.5th percentile with a load, so no single claim sets a rating factor.
"Blend recipes: weight of each model family"	We built two pipelines. Oussama's models frequency and severity with a GLM selected by BIC and a spline GAM, then averages them with LightGBM and CatBoost.
(same)	Optional, 8 seconds: Severity keeps only three or four variables: claim size is mostly chance, so the rating signal is in frequency.
(same)	Yichuan's fits five Tweedie models on pure premium, and blend weights must beat equal weights on both time folds.
"Public D2 by submission"	Both validate on a later year, never a random split.
(same)	The leaderboard caught one gap validation missed: 2025 claim history is incomplete, so we count only claims known on 1 January.
"Standing on both boards"	S09, our ranked entry, scored 0.13689 privately. Blended 70/30 with Oussama's Blend4, it reaches 0.13718, as a late submission.

It needs about 50 seconds cut elsewhere, since the team script is already full. If the Overview keeps its line on S09, drop the S09 sentence in the last row.

Version B: drop-in for the team script (94 words, about 0:42)

It replaces the five Part 1 lines in section 1, from "Yichuan's approach blends five Tweedie models" to "capped at the 99.5th percentile", so the total stays at 648 words. The first three lines go over "Blend recipes", the last two over "Data quality: ten documented issues", as now.

Spoken (English)
Oussama models frequency times severity with a BIC-selected GLM and a GAM, averaged with LightGBM and CatBoost.
Yichuan blends five Tweedie models on pure premium; weights must beat equal weights on both time folds.
The 0.7 to 0.3 blend of the two scored 0.13718 privately, a late submission, not ranked.
Claims stop at the end of 2024: only 77 percent of 2024 motor cost is reported, and bodily injury is 14 percent reported at 30 days.
So claims are developed with a 2021 reporting curve and capped at the 99.5th percentile with a load.

What it gains over the current lines: the 77% figure that makes development concrete, and the frequency × severity structure. What it drops: "claims match to the policy row containing the loss date", the least telling line. Whoever speaks can say "my pipeline" instead of their own name.